MILITARY SPECIFICATION

CU-2113( )/SRC COUPLER-ISOLATOR

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers CU-2113( )/SRC Coupler-Isolators which enable the simultaneous use of a shipboard 2 megahertz (MHz) to 6 MHz broadband transmitting antenna as a 10 kilohertz (kHz) to 30 MHz receiving antenna.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue listed in the Department of Defense Index of Specifications and Standards (DoDISS) and its supplements, form a part of this document to the extent specified herein. The date of the applicable DoDISS and supplements thereto shall be as specified in the solicitation.

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-S-901 | Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment And Systems Requirements For |
| MIL-P-15024 | Plates, Tags And Bands For Identification Of Equipment |
| MIL-E-16400 | Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For |
| MIL-E-17555 | Electronic And Electrical Equipment, Accessories, And Repair Parts; Packaging And Packing Of |
| MIL-E-21981 | Electronics Type Designations, Identification Plates And Markings, Requirements For |
| MIL-C-23183 | Capacitors, Fixed Or Variable, Vacuum Dielectric, General Specification For |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-109 | Quality Assurance Terms And Definitions |
| MIL-STD-167-1 | Mechanical Vibrations Of Shipboard Equipment (Type I - Environmental And Type II - Internally Excited) |
| MIL-STD-454 | Standard General Requirements For Electronic Equipment |
| MIL-STD-461 | Electromagnetic Interference Characteristics Requirements For Equipment |
| MIL-STD-462 | Electromagnetic Interference Characteristics, Measurement Of |
| MIL-STD-810 | Environmental Test Methods |

**Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-5043), Washington, D.C. 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.**

FSC 5825

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

DRAWINGS

MILITARY

NAVAL ELECTRONIC SYSTEMS COMMAND
73E9N100 Coupler-Isolator CU-2113/SRC Final Assembly

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The CU-2113( )/SRC Coupler-Isolator to be supplied shall be in accordance with Naval Electronic Systems Command drawing 73E9N100 except as hereinafter specified.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.3). The first article shall be completely representative of the production units.

3.3 Performance requirements.

3.3.1 TRANSMITTER port to ANTENNA port insertion loss. The CU-2113( )/SRC Coupler-Isolator insertion loss with respect to a 50 ohm system from the TRANSMITTER port to the ANTENNA port with a 50 ohm (non-reactive) resistive load on each the ANTENNA, HF RECEIVE, and LF-MF RECEIVE ports when measured at each frequency from 2 MHz to 6 MHz inclusive shall be less than 0.5 decibels (dB).

3.3.2 ANTENNA port to HF RECEIVE port insertion loss. The CU-2113( )/SRC Coupler-Isolator insertion loss with respect to a 50 ohm system from the ANTENNA port to the HF RECEIVE port with a short circuit on the TRANSMITTER port and a 50 ohm (non-reactive) resistive load on each the LF-MF RECEIVE and HF RECEIVE ports when measured at frequencies from 2 MHz to 30 MHz inclusive shall be within the limits shown in FIGURE 1. The CU-2113( )/SRC insertion loss with respect to a 50 ohm system from the ANTENNA port to the HF RECEIVE port with an open circuit on the TRANSMITTER port and a 50 ohm (non-reactive) resistive load on each the LF-MF RECEIVE and HF RECEIVE ports when measured at frequencies from 2 MHz to 30 MHz inclusive shall be within the limits shown in FIGURE 2.

3.3.3 ANTENNA port to LF-MF RECEIVE port insertion loss. The CU-2113( )/SRC Coupler-Isolator insertion loss with respect to a 50 ohm system from the ANTENNA port to the LF-MF RECEIVE port with a 4720 picofarad (pF) vacuum capacitor conforming to MIL-C-23183, on the TRANSMITTER port and a 50 ohm (non-reactive) resistive load on each the HF RECEIVE and LF-MF RECEIVE ports when measured at frequencies from 10 kHz to 2 MHz inclusive shall be within the limits shown in FIGURE 3.

3.3.4 TRANSMITTER port to HF RECEIVE port insertion loss. The CU-2113( )/SRC Coupler-Isolator insertion loss with respect to a 50 ohm system from the TRANSMITTER port to the HF RECEIVE port with a 50 ohm (non-reactive) resistive load on each the ANTENNA, HF RECEIVE, and LF-MF RECEIVE ports when measured at frequencies from 2 MHz to 6 MHz inclusive shall be not less than the limits shown in FIGURE 4. The CU-2113( )/SRC insertion loss with respect to a 50 ohm system from the TRANSMITTER port to the HF RECEIVE port with a 16.7 ohm (non-reactive) resistive load on the ANTENNA port and a 50 ohm (non-reactive) resistive load on each the HF RECEIVE and LF-MF RECEIVE port when measured at frequencies from 2 MHz to 6 MHz inclusive shall be not less than the limits shown in FIGURE 5. The CU-2113( )/SRC insertion loss with respect to a 50 ohm system from the TRANSMITTER port to the HF RECEIVE port with a 150 ohm (non-reactive) resistive load on the ANTENNA port and a 50 ohm (non-reactive) resistive load on each the LF-MF RECEIVE and HF RECEIVE ports when measured at frequencies from 2 MHz to 6 MHz inclusive shall be not less than the limits shown in FIGURE 6."



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

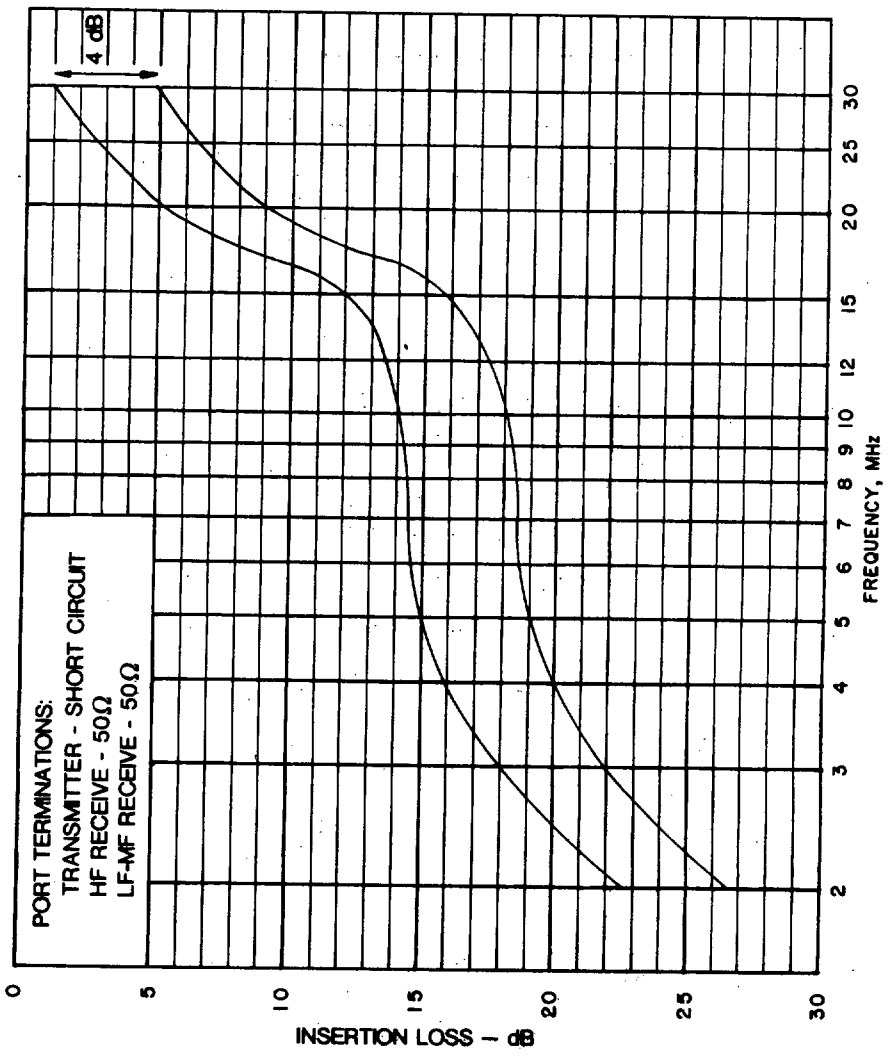



FIGURE 1. CU-2113( )/SRC insertion loss limits with respect to 50Ω system — ANTENNA port to HF RECEIVE port. TRANSMITTER — short circuit.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

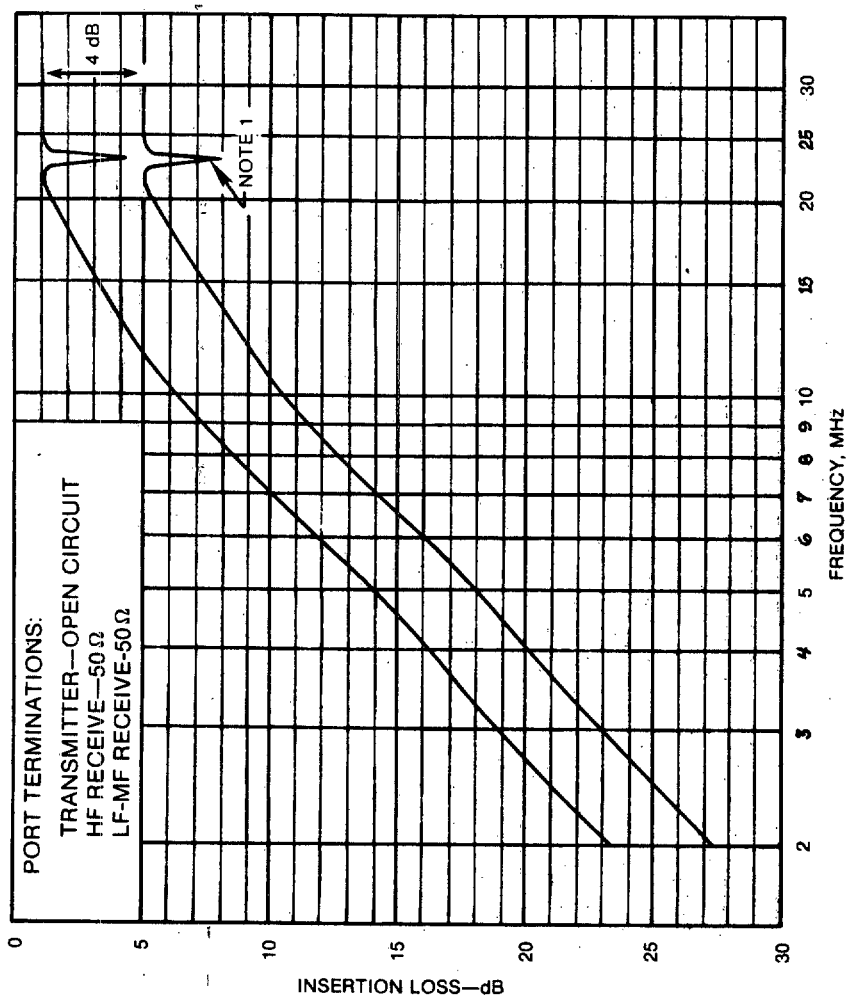



FIGURE 2. CU-2113( )/SRC insertion loss limits with respect to 50 Ω system—ANTENNA port to HF RECEIVE port TRANSMITTER—open circuit.

NOTES:
1. Frequency corresponding to the maximum insertion loss of the notch in the insertion loss response shall be between 23.0 MHz and 24.75 MHz.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

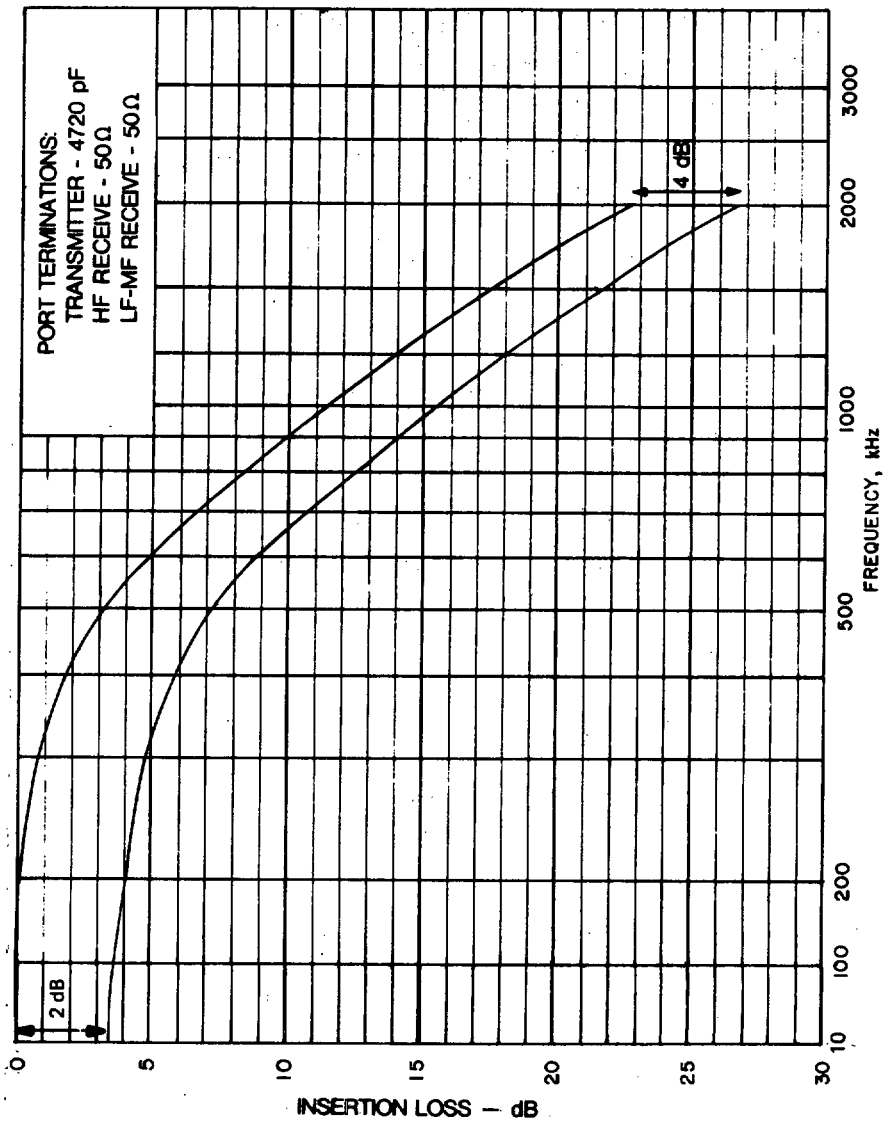



FIGURE 3. CU-2113( )/SRC insertion loss limits with respect to 50Ω system—ANTENNA port to LF-MF RECEIVE port.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

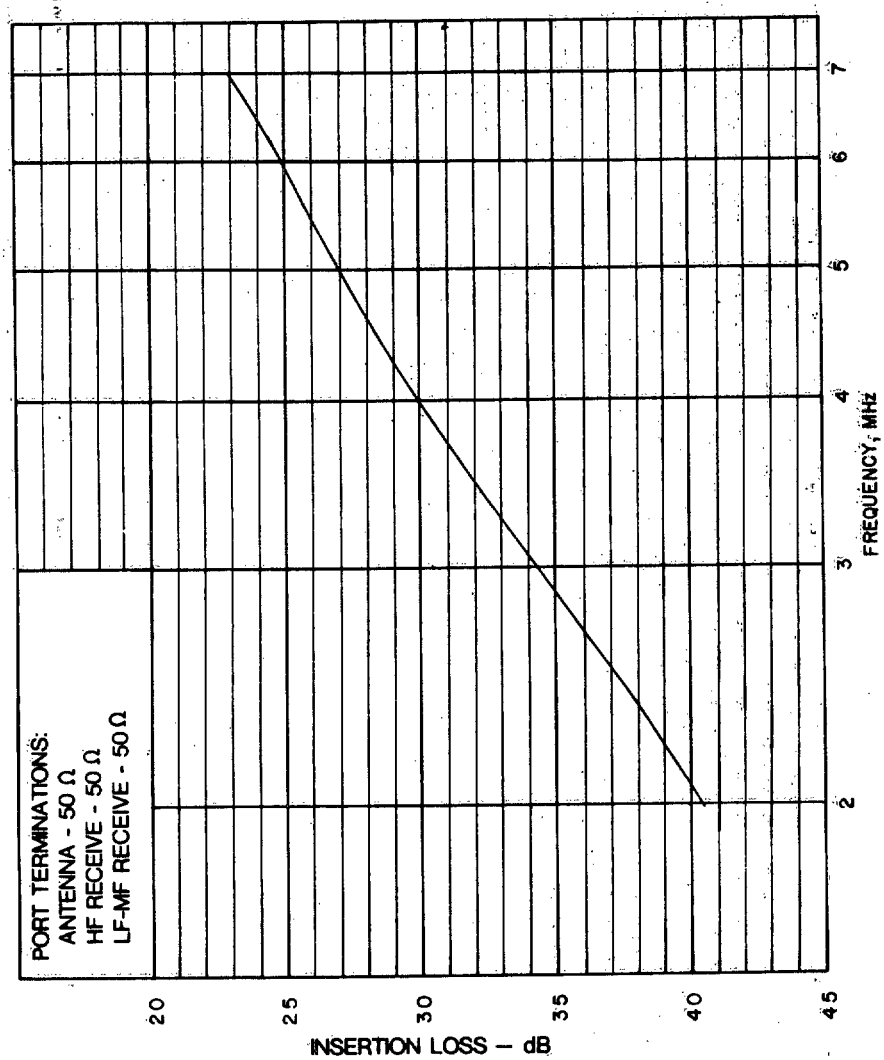



FIGURE 4. CU-2113( )/SRC insertion loss limits with respect to 50 Ω system— TRANSMITTER port to HF RECEIVE port, ANTENNA 50 Ω.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

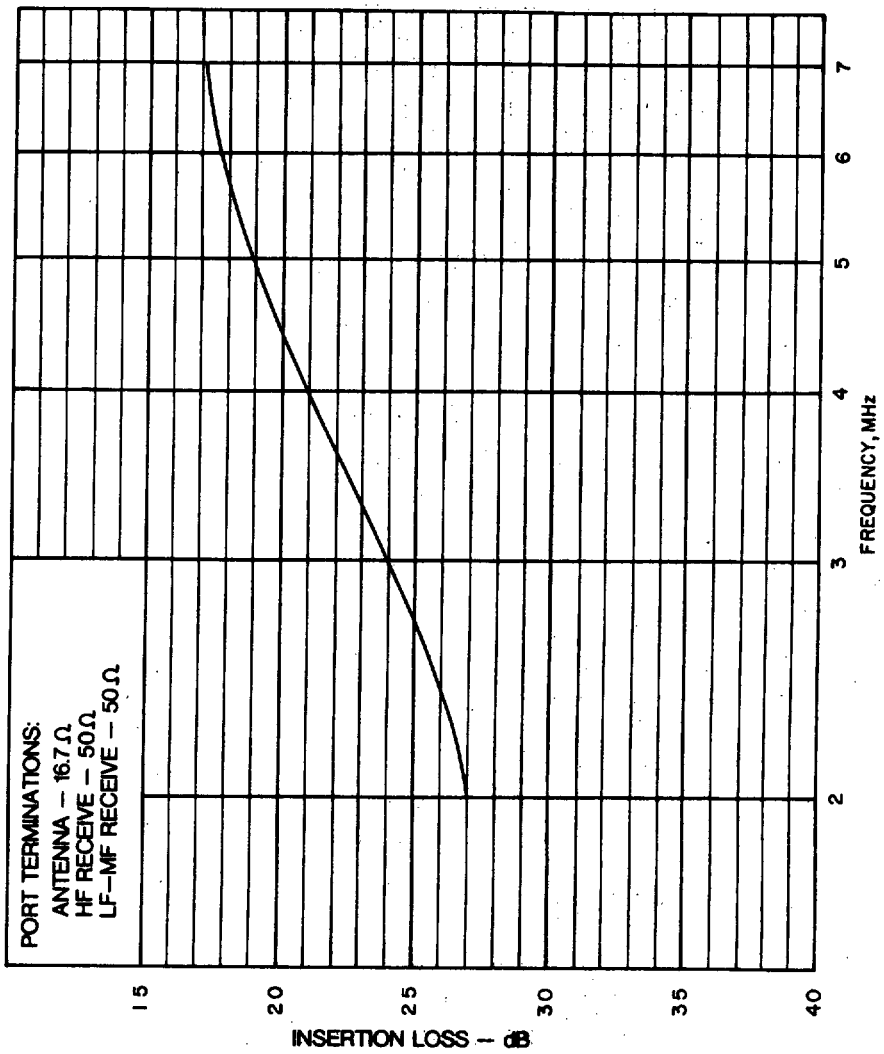



FIGURE 5. CU–2113( )/SRC insertion loss limits with respect to 50 Ω. system – TRANSMITTER port to HF RECEIVE port. ANTENNA 16.7 Ω.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

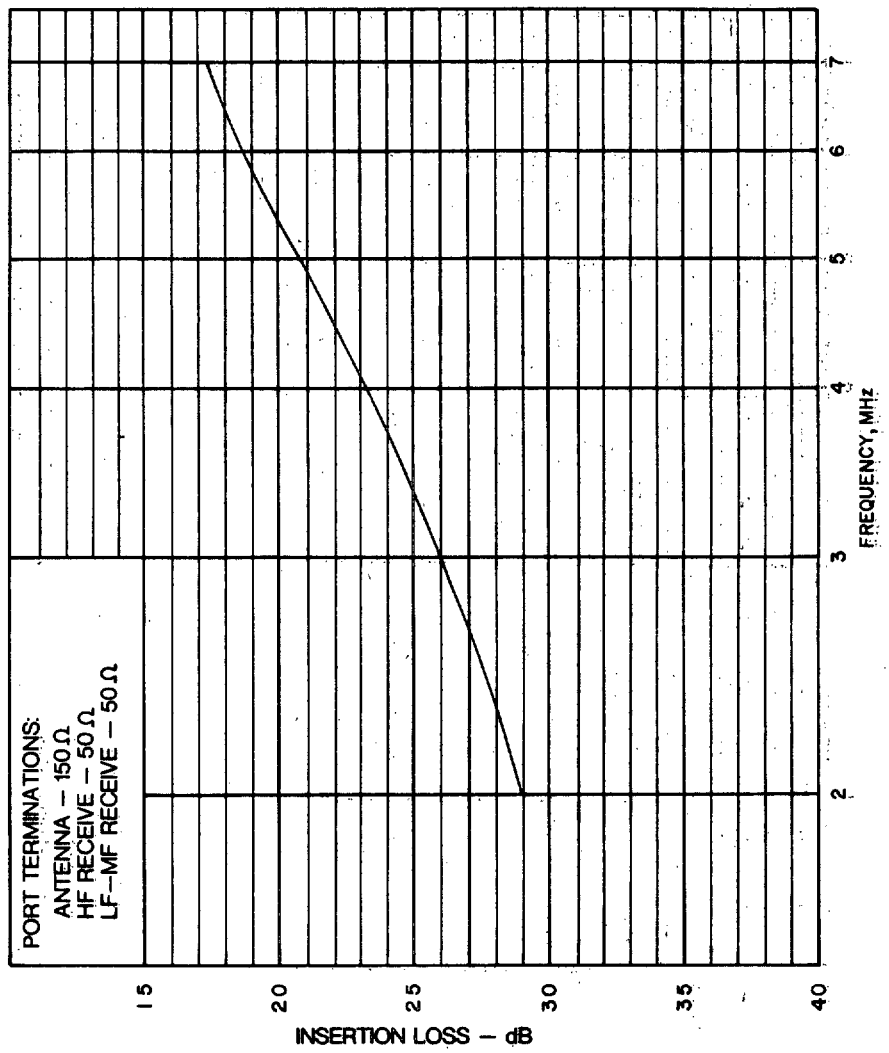



FIGURE 6. CU–2113( )/SRC insertion loss limits with respect to 50 Ω system – TRANSMITTER port to HF RECEIVE port; ANTENNA 150 Ω.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

3.3.5 Radio frequency input power. The CU-2113( )/SRC Coupler-Isolator shall operate without damage with a total maximum radio frequency average power of 5.04 kilowatts (kW) applied to the TRANSMITTER port of the CU-2113( )/SRC at frequencies in the range from 2.0 MHz through 6.0 MHz. The total heat dissipation to the surrounding air from the CU-2113( )/SRC with the foregoing 5.04 kW applied to the TRANSMITTER port shall not exceed 200 watts (W).

3.3.6 Generation of intermodulation products. Intermodulation product levels generated in the CU-2113( )/SRC Coupler-Isolator shall each be less than or equal to 10 dB above one microvolt ($\mu$V) at the HF RECEIVE port of the CU-2113( )/SRC when: the ANTENNA, HF RECEIVE, and LF-MF RECEIVE ports of the CU-2113( )/SRC are each terminated in a 50 ohm (non-reactive) resistive load; 500 W (average power) at each of two different frequencies in the 2.0 MHz through 6.0 MHz frequency range are applied to the TRANSMITTER port of the CU-2113( )/SRC.

3.4 Environmental service conditions. The CU-2113( )/SRC Coupler-Isolator shall be capable of continuous reliable operation in accordance with the performance requirements of this specification under the environmental conditions specified herein.

3.4.1 Temperature. The CU-2113( )/SRC Coupler-Isolator shall maintain specified performance when subjected to temperatures throughout the complete operating temperature range 4 of MIL-E-16400. The CU-2113( )/SRC shall not be damaged nor shall performance be degraded when restored to the operating temperature range 4 of MIL-E-16400 after being subjected to the complete non-operating temperature range 4 of MIL-E-16400.

3.4.2 Humidity. The CU-2113( )/SRC Coupler-Isolator shall maintain specified performance when subjected to relative humidities ranging from 0 to 95 percent for both continuous and intermittent periods.

3.4.3 Salt fog (spray). Salt fog (spray) shall be in accordance with the Salt fog (spray) paragraph of MIL-E-16400 for sheltered equipment.

3.4.4 Shock. The CU-2113( )/SRC Coupler-Isolator shall be capable of withstanding the grade A, type A, class I shock test of MIL-S-901 without impairment which would cause CU-2113( )/SRC to malfunction or which would cause degradation in specified performance after completion of the test. Degradation in specified performance during the shock test impact will be allowed.

3.4.5 Vibration. The CU-2113( )/SRC Coupler-Isolator shall maintain specified performance when subjected to the type I vibration test of MIL-STD-167-1.

3.4.6 Inclination. The CU-2113( )/SRC Coupler-Isolator shall maintain specified performance when subjected to inclination angles of up to and including 45 degrees from the horizontal.

3.5 Electromagnetic compatibility. The CU-2113( )/SRC Coupler-Isolator shall be self-compatible and shall conform to the requirements of MIL-STD-461 for class 1A and 1B equipments as follows:

a. RE02: MIL-STD-461 limits shall be increased as 10 $\log_{10}$ (P/(1 watt)) for all values of RF input power, P, over one watt.
b. RS03: MIL-STD-461 limits unchanged except that levels shall be 50 volts/meter from 2.0 MHz through 30.0 MHz.

During susceptibility tests the performance of the Coupler-Isolator shall not deviate from the requirements of 3.3.

3.6 Identification and marking. An identification plate conforming to the requirements of MIL-E-21981 and which indicates the equipment nomenclature and serial number shall be attached to each CU-2113( )/SRC Coupler-Isolator offered for delivery. Adhesive backed identification plates, type G of MIL-P-15024, shall not be used.

3.7 Workmanship. Workmanship shall conform to Naval Electronic Systems Command drawing 73E9N100 and shall comply with requirement 9 of MIL-STD-454.

3.7.1 Workmanship screen. All equipment shall withstand a defect detection vibration screen of random type vibration at 0.04$g^2$/Hz ($\pm$3 dB) from 80 to 350 Hz, and temperature cycling to the temperature range specified in 3.4.1.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) Surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.1.3 Standard test conditions. Unless otherwise specified in the individual test methods (see 4.5), all measurements and tests shall be made under the following ambient conditions:

a. Temperature $23^{\circ} \pm 10^{\circ}C$ ($73^{\circ} \pm 18^{\circ}F$)
b. Relative humidity $50 \begin{smallmatrix} + 25 \\ - 10 \end{smallmatrix}$ percent
c. Atmospheric pressure $7? \begin{smallmatrix} + 50 \\ - 75 \end{smallmatrix}$ mm of mercury ($28.5 \begin{smallmatrix} + 2.0 \\ - 3.0 \end{smallmatrix}$ inches of mercury)

4.2 Classification of inspection. The method of examination and testing shall be classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
c. Inspection of preparation for delivery (see 4.6)

4.3 First article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the inspections so identified and listed in TABLE I. The number of first article CU-2113( )/SRC Coupler-Isolators required for first article inspection shall be as specified in the contract (see 6.2 and 6.3).

4.4 Quality conformance inspection. Quality conformance inspection shall be made on each Coupler-Isolator offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process, such as functional and performance tests, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Quality conformance inspection shall include the tests so indicated in TABLE I.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

TABLE I. CU-2113( )/SRC Coupler-Isolator inspection and test requirements.

| Inspection or test | Requirement paragraph | Test method paragraph | First article inspection | Quality conformance inspection |
|---|---|---|---|---|
| Surface examination | 3.1,3.6,3.7 | 4.5.1 | X | X |
| Workmanship screen | 3.7.1 | 4.5.2 | X | X |
| Transmitter port to ANTENNA port insertion loss test | 3.3.1 | 4.5.3.1 | X | X |
| ANTENNA port to HF RECEIVE port insertion loss test | 3.3.2 | 4.5.3.2 | X | X |
| ANTENNA port to LF-MF RECEIVE port insertion loss test | 3.3.3 | 4.5.3.3 | X | X |
| TRANSMITTER port to HF RECEIVE port insertion loss test | 3.3.4 | 4.5.3.4 | X | X |
| RF input power test | 3.3.5 | 4.5.4 | X | |
| Intermodulation test | 3.3.6 | 4.5.5 | X | |
| Temperature test | 3.4.1 | 4.5.6 | X | |
| Humidity test | 3.4.2 | 4.5.7 | X | |
| Salt fog test | 3.4.3 | 4.5.8 | X | |
| Shock test | 3.4.4 | 4.5.9 | X | |
| Vibration test | 3.4.5 | 4.5.10 | X | |
| Inclination test | 3.4.6 | 4.5.11 | X | |
| EMC test | 3.5 | 4.5.12 | X | |

4.4.1 Nonconforming quality conformance inspection units. If a Coupler-Isolator fails any of the inspections listed for quality conformance inspection in TABLE I, the contractor shall immediately investigate the cause of failure; shall report to the Government inspector and the Government inspection activity the failure and the results of the investigation thereof; and shall take corrective action to ensure the failed Coupler-Isolator will conform to the requirements of this specification. The contractor shall then institute corrective action for Coupler-Isolators which were manufactured both previously and subsequentially with the same conditions, materials, processes, and so forth, as the failed unit to ensure the problem which caused the failure will be eliminated in all delivered Coupler-Isolators.

4.5 Test methods.

4.5.1 Surface examination. The CU-2113( )/SRC Coupler-Isolator and its associated parts, components, subassemblies, and assemblies shall be inspected for conformance to Naval Electronic Systems Command drawing 73E9N100 and to verify compliance with the requirements of this specification. The Coupler-Isolator shall be examined for conformance to requirements for, interalia, the following: dimensions, parts and materials, interconnections, finish, identification and marking, safety, and workmanship. In the process of examination the Coupler-Isolator shall not be disassembled in a manner that will affect the performance, durability, or appearance of the unit.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

4.5.2 Workmanship screen. Prior to conducting temperature cycling test, vibration shall be performed on each equipment. All the hardware, including cables and connectors, shall be exposed to vibration.

4.5.2.1 Random vibration. The vibration shall be random, or, subject to Procuring activity approval, pseudorandom or complex wave-form vibration for an accumulated time of ten minutes in the test item axis deemed most susceptible to vibration excitation. All test items shall be hard mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 7. The control accelerometer shall be located next to one of the mounting points of the item under test. The test item shall be energized during vibration and appropriate input signals applied to observe any abnormal conditions of the output functional characteristics. Any failure occurring during the test shall require corrective action before the test is resumed.

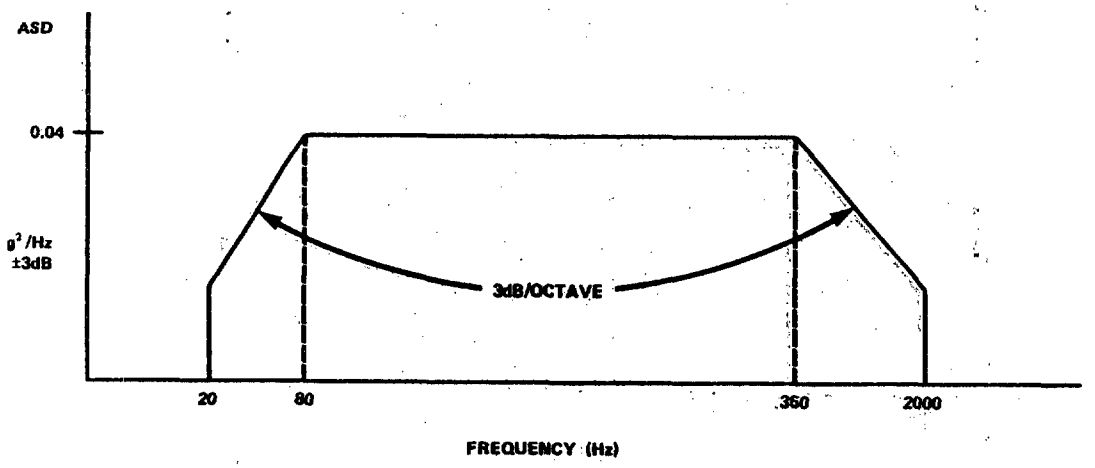



FIGURE 7. Random vibration curve.

NOTE: The acceleration spectral density (ASD) shall be controlled by an accelerometer signal with a filtered bandwidth of 10 Hz up to a vibration frequency of 500 Hz and a 100 Hz bandwidth above 500 Hz.

4.5.2.2 Temperature cycling. Each equipment shall be subjected to the temperature curve indicated below. The number of cycles required shall be determined by the electrical and electronic parts count and the parts count chart below. The temperature rate of change shall be not less than 5° Celsius (C) per minute. The equipment enclosure shall be opened for maximum exposure to the changing temperature. The last temperature cycle shall be failure free. A functional operating test shall be performed. The dwell time shall be 95 percent of the time required for the largest electrical or electronic part to become temperature stabilized. Temperature stabilization shall be as defined in MIL-STD-810 for the non-operating condition.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

| Electrical or Electronic parts count | No. of temperature cycles |
|---|---|
| 500 or less | 3 |
| 501 to 1000 | 4 |
| 1001 to 2000 | 6 |
| 2001 to 3000 | 8 |
| 3001 and up | 10 |

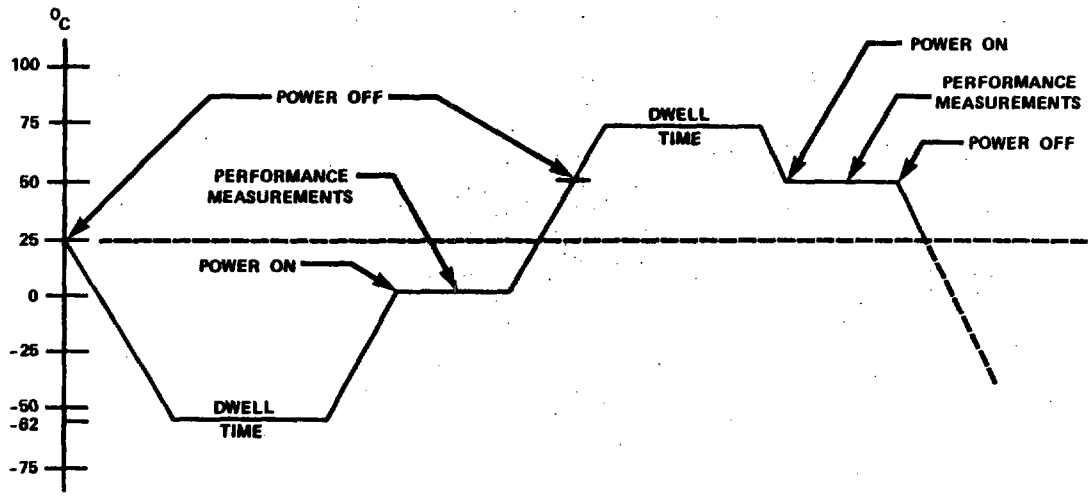



FIGURE 8. Temperature curve (one cycle).

4.5.3 Functional test.

4.5.3.1 TRANSMITTER port to ANTENNA port insertion loss test. Swept frequency measurements of the insertion loss (with respect to a 50 ohm system) from the TRANSMITTER port to the ANTENNA port of the Coupler-Isolator shall be made. The ANTENNA, HF RECEIVE, and LF-MF RECEIVE ports shall each be terminated in a 50 ohm (non-reactive) resistive load and attenuated (10 dB minimum line attenuation) 50 ohm source impedance swept frequency signals in the 2.0 MHz to 8.0 MHz minimum frequency range applied to the TRANSMITTER port for these measurements. Calibrated X-Y plots (linear in frequency and logarithmic in insertion loss) shall be obtained and the results compared with the requirements of 3.3.1.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

4.5.3.2 ANTENNA port to HF RECEIVE port insertion loss test. Swept frequency measurements of the insertion loss (with respect to a 50 ohm system) from the ANTENNA port to the HF RECEIVE port of the Coupler-Isolator shall be made. Attenuated (10 dB minimum line attenuation) 50 ohm source impedance swept frequency signals in the 2.0 MHz to 30.0 MHz minimum frequency range shall be applied to the ANTENNA port for these measurements with the HF RECEIVE and LF-MF RECEIVE ports each terminated in a 50 ohm (non-reactive) resistive load and the TRANSMITTER port terminated in a short circuit for the first series of measurements and in an open circuit for the second. Calibrated X-Y plots (linear in frequency and logarithmic in insertion loss) shall be obtained and the results compared with the requirements of 3.3.2.

4.5.3.3 ANTENNA port to LF-MF RECEIVE port insertion loss test. Swept frequency measurement of the insertion loss (with respect to a 50 ohm system) from the ANTENNA port to the LF-MF RECEIVE port of the Coupler-Isolator shall be made. Attenuated (10 dB minimum line attenuation) 50 ohm source impedance swept frequency signals in the 10 kilohertz (kHz) to 2.0 MHz minimum frequency range shall be applied to the ANTENNA port for these measurements with the LF-MF RECEIVE and HF RECEIVE ports each terminated in a 50 ohm (non-reactive) resistive load and the TRANSMITTER port terminated in a vacuum capacitor conforming to MIL-C-23183 such that the total external capacitance at the TRANSMITTER port is equal to 4720 pF (± 3 percent). Calibrated X-Y plots (linear in frequency and logarithmic in insertion loss) shall be obtained and the results compared with the requirements of 3.3.3.

4.5.3.4 TRANSMITTER port to HF RECEIVE port insertion loss test. Swept frequency measurements of the insertion loss (with respect to a 50 ohm system) from the TRANSMITTER port to the HF RECEIVE port of the Coupler-Isolator shall be made. Attenuated (10 dB minimum line attenuation) 50 ohm source impedance swept frequency signals in the 2.0 MHz to 8.0 MHz minimum frequency range shall be applied to the TRANSMITTER port for these measurements with the HF RECEIVE and LF-MF RECEIVE ports each terminated in a 50 ohm (non-reactive) resistive load and the ANTENNA port terminated, in turn, in a 16.7 ohm, 50 ohm, and 150 ohm (non-reactive) resistive load (series of three measurements). Calibrated X-Y plots (linear in frequency and logarithmic in insertion loss) shall be obtained and the results compared with the requirements of 3.3.4.

4.5.4 RF input power test. An average power of 5.04 kilowatts shall be applied to the TRANSMITTER port of the Coupler-Isolator at a single frequency or a multiple frequencies (5.04 kW total input power from all sources) in the frequency range of from 2.0 MHz to 6.0 MHz with the ANTENNA, HF RECEIVE, and LF-MF RECEIVE ports each terminated in a 50 ohm (non-reactive) resistive load. Temperature indicating labels or paints or both shall be affixed to internal components of the Coupler-Isolator to record the maximum temperatures reached during the test. The ambient temperature surrounding the Coupler-Isolator shall be maintained at 50 degrees C (± 2 degrees C) during the test. RF power shall be applied to the Coupler-Isolator until the internal temperatures of the components stabilize but for a time period not less than four hours. Bridge measurements of the direct current (DC) resistance across the HF RECEIVE port terminals shall be made both prior to and after the test (after return of Coupler-Isolator temperature to room ambient temperature). The ANTENNA port to HF RECEIVE port insertion loss test of 4.5.3.2 shall be made on the Coupler-Isolator both prior to and after the test (after return of the Coupler-Isolator temperature to room ambient temperature). The temperatures reached by the components in the Coupler-Isolator during the test shall be compared with the temperature ratings of the components; the before and after test HF RECEIVE port DC resistance measurements and the ANTENNA port to HF RECEIVE port insertion loss test results shall also be compared. The Coupler-Isolator shall not sustain damage during the test.

4.5.5 Intermodulation test. Five hundred watts (average power) at each of two different frequencies in the frequency range of 2.0 MHz to 6.0 MHz shall be applied to the TRANSMITTER port of the Coupler-Isolator and the levels of the intermodulation products generated therein by these frequencies and appearing at the HF RECEIVE port shall be determined. The LF-MF RECEIVE and ANTENNA ports shall each be terminated in 50 ohm (non-reactive) resistive loads for these measurements. The HF RECEIVE port shall be terminated in an equivalent 50 ohm resistive load for determination of the levels of the intermodulation products.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

4.5.6 Temperature test. The temperature test shall be in accordance with the method 502 low temperature test of MIL-STD-810 and the method 501, procedure I, high temperature test of MIL-STD-810 except as otherwise specified herein. In the method 502 low temperature test of MIL-STD-810:

a. STEP 2 - Storage or non-operating temperature shall be minus 62 degrees Celsius, +0 degrees C, -5 degrees C, for a duration of 24 hours.
b. STEP 4 - Temperature shall be 0 degrees C.
c. STEP 5 - The ANTENNA port to HF RECEIVE port insertion loss test of 4.5.3.2 shall be conducted on the Coupler-Isolator; this test shall constitute operation of the test item.
d. STEP 7 - The ANTENNA port to HF RECEIVE port insertion loss test of 4.5.3.2 shall be conducted on the Coupler-Isolator; this test shall constitute operation of the test item.

In the method 501, procedure I, high temperature test of MIL-STD-810:

a. STEP 4 - Temperature shall be 50 degrees C.
b. STEP 5 - The ANTENNA port to HF RECEIVE port insertion loss test of 4.5.3.2 shall be conducted on the Coupler-Isolator; this test shall constitute operation of the test item.
c. STEP 7 - The ANTENNA port to HF RECEIVE port insertion loss test of 4.5.3.2 shall be conducted on the Coupler-Isolator; this test shall constitute operation of the test item.

The ANTENNA port to HF RECEIVE port insertion loss test of 4.5.3.2 shall be the pretest performance check, performance check during test, post-test performance check, and operation of the item specified in MIL-STD-810. The ANTENNA port to HF RECEIVE port insertion loss test of 4.5.3.2, however, need only be conducted on the Coupler-Isolator a total of three times for each the low temperature and the high temperature tests of MIL-STD-810. Damage to the Coupler-Isolator or failure of the ANTENNA port to HF RECEIVE port insertion loss test data obtained during and post-test to duplicate pretest data (or both) shall constitute failure of the Coupler-Isolator to conform to the requirements of this specification.

4.5.7 Humidity test. The humidity test shall be in accordance with the method 507, procedure IV, humidity test of MIL-STD-810 except as otherwise specified herein. In the method 507, procedure IV, humidity test of MIL-STD-810:

a. STEP 4 - The ANTENNA port to HF RECEIVE port insertion loss test of 4.5.3.2 shall be conducted on the Coupler-Isolator.
b. STEP 5 - Change the maximum temperature of the figure 507.1-3 procedure IV humidity cycle during the normal 48 to 64 cumulative hours time period from 60 degrees C to 50 degrees C. The ANTENNA port to HF RECEIVE port insertion loss test of 4.5.3.2 shall be conducted on the Coupler-Isolator during the second cycle at 50 degrees C (+0,-3 degrees C) immediately prior to decreasing to 30 degrees C (± 5 degrees C).
c. STEP 6 - The ANTENNA port to HF RECEIVE port insertion loss test of 4.5.3.2 shall be conducted on the Coupler-Isolator.
d. STEP 8 - The ANTENNA port to HF RECEIVE port insertion loss test of 4.5.3.2 shall be conducted on the Coupler-Isolator.

The ANTENNA port to HF RECEIVE port insertion loss test of 4.5.3.2 shall be the pretest required. Damage to the Coupler-Isolator (including degradation of finishes, and so forth) or failure of the ANTENNA port to HF RECEIVE port insertion loss data taken during test and post test to duplicate pretest data (or both) shall constitute failure of the Coupler-Isolator to conform to the requirements of this specification.

4.5.8 Salt fog test. The salt fog test for sheltered equipment shall be conducted on the Coupler-Isolator in accordance with MIL-E-16400. Physical degradation of the items subjected to the test shall constitute failure of the Coupler-Isolator to conform to specified requirements.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

4.5.9 Shock test. The Coupler-Isolator shall be shock tested in accordance with the grade A, type A, class I shock test requirements of MIL-S-901. Simulated loads, such as weights or other dummy masses, shall not be used. The test shall be performed with the normal assemblies and units in their functionally operating positions. The insertion loss tests 4.5.3.1 through 4.5.3.4 shall be conducted on the Coupler-Isolator both prior to and after the impact of the shock test. Excessive damage as specified in MIL-S-901 or failure of the insertion loss test data after the shock test to duplicate insertion loss test data prior to the shock test (or both) shall constitute failure of the Coupler-Isolator to conform to specified requirements.

4.5.10 Vibration test. The Coupler-Isolator shall be subjected to the type I vibration test of MIL-STD-167-1 with an upper vibration limit of 50 cycles per second. Simulated loads, such as weights or other dummy masses, shall not be used. The test shall be performed with the normal assemblies and units in their functionally operating positions. The insertion loss tests of 4.5.3.1 through 4.5.3.4 shall be conducted on the Coupler-Isolator both prior to and after the vibration testing. The insertion loss test of 4.5.3.2 (ANTENNA port to HF RECEIVE port insertion loss test) shall be conducted on the Coupler-Isolator during the vibration testing to determine effects on insertion loss. Damage, degradation of performance, or failure of the insertion loss test data after vibration testing to duplicate the insertion loss test data prior to vibration testing (or combinations thereof) shall constitute failure of the Coupler-Isolator to conform to specified requirements.

4.5.11 Inclination test. The Coupler-Isolator shall be subjected to the inclination test for surface ship installations of MIL-E-16400 except as otherwise specified herein. The insertion loss test of 4.5.3.2 (ANTENNA port to HF RECEIVE port insertion loss test) shall be conducted on the Coupler-Isolator during the inclination test to determine effects on insertion loss; this test shall constitute operation of the equipment for the test. The insertion loss tests of 4.5.3.1 through 4.5.3.4 shall be conducted on the Coupler-Isolator both prior to and after the inclination testing. Damage, degradation of performance, or failure of the insertion loss test data after inclination testing to duplicate the insertion loss test data prior to vibration testing (or combinations thereof) shall constitute failure of the Coupler-Isolator to conform to specified requirements.

4.5.12 EMC test. The RE02 and RS03 electromagnetic interference emission and susceptibility tests shall be conducted on the Coupler-Isolator in accordance with MIL-STD-462 except as otherwise specified herein (see also 3.5). Solid shield coaxial cable shall be used for all test connections to the TRANSMITTER and ANTENNA ports of the Coupler-Isolator. Double shielded braid (RG-214/U or equivalent) coaxial cable shall be used for each test connection requiring a coaxial cable to the HF RECEIVE and LF-MF RECEIVE ports of the Coupler-Isolator.

4.5.12.1 RE02 test. For the RE02 test the ANTENNA port of the Coupler-Isolator shall be terminated in a 50 ohm (non-reactive) resistive load with the HF RECEIVE and LF-MF RECEIVE ports terminated in the short circuit terminations supplied with the Coupler-Isolator. The radiated emissions from the Coupler-Isolator (not to include the effects of coaxial cables and connectors attached to the Coupler-Isolator) shall be determined for each of five applications of continuous wave 1000 watt signals from a 50 ohm source impedance applied to the TRANSMITTER port of the Coupler-Isolator; the five signals applied to the TRANSMITTER port shall be at 2.0 MHz, 3.0 MHz, 4.0 MHz, 5.0 MHz, and 6.0 MHz. The radiated emissions at the 2.0 MHz, 3.0 MHz, 4.0 MHz, 5.0 MHz, and 6.0 MHz frequencies (only) when each is applied to the TRANSMITTER port of the Coupler-Isolator shall each be derated by the factor indicated in 3.5.a to determine compliance with MIL-STD-461 RE02 test requirements.

4.5.12.2 RS03 test. For the RS03 test the radiated susceptibility shall be determined at the ANTENNA, HF RECEIVE, and LF-MF RECEIVE ports only. The ANTENNA, HF RECEIVE, LF-MF RECEIVE, and TRANSMITTER ports shall each be terminated in a 50 ohm (non-reactive) resistive load for this test.

4.5.12.3 Failure. Degradation of performance or failure of the Coupler-Isolator to comply with the applicable requirements of MIL-STD-461 (or both) shall constitute failure of the Coupler-Isolator to conform to specified requirements.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

4.6 Inspection of preparation for delivery. Inspection of preparation for delivery shall be conducted on every Coupler-Isolator assigned for shipment. Inspection of preparation for delivery shall consist of inspections to verify compliance with the requirements of section 5. This inspection shall be successfully concluded prior to securing the container for shipment.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing and marking. Unless otherwise specified, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing and marking specified in MIL-E-17555 (see 6.2)

6. NOTES AND CONCLUDING MATERIAL

6.1 Intended use. This equipment is intended for sheltered U.S. Navy shipboard installation.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. First article sample and quantity required (see 3.2, 4.3).
c. Levels of preservation, packaging, packing and marking required (see 5.1).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation (DAR). The first article should be a first production item. The first article should consist of the number of units specified in the contract. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, test, and approval of the first article.

Preparing activity:
NAVY-EC
(Project 5825-N012(EC))



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-C-28855(EC)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1426 1 OCT 76

EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D.C. 20360

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
NAVY DEPARTMENT
DOD 316

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D.C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:53Z
Check the source to verify that this is the current version before use.